Nondimensional Staged Feedback Control of Mitochondrial Heteroplasmy Dynamics with Host-Emulated Verification on an Ultra-Low-Power RV32IMF_Zve32x Neural Processing Unit

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Abstract

Restoring bioenergetic function in cells burdened by mutant or damaged mitochondrial populations (D) alongside wild-type populations (W) requires navigating a nonlinear coupling between clearance bottlenecks, reactive oxygen species (ROS) damage, and replicative selection. When upstream proteostatic or environmental bottlenecks (B) elevate energetic impedance, premature biogenesis stimulation (u_2) amplifies electron-transport back-pressure and accelerates mutant expansion when D possesses a replicative or survival advantage ($k_{\mathrm{rep,adv}} \geq 1$). Here we formulate an eight-state ordinary differential equation (ODE) model of coupled mitochondrial quality control and analyze a three-input staged state-feedback threshold control law: Stage 0 upstream bottleneck clearance (u_0), Stage 1 selective mitophagic clearance (u_1), and Stage 2 biogenesis induction (u_2) gated at $h(t) = D(t)/(W(t) + D(t)) \leq h_{\mathrm{gate}} = 0.080$ and $B(t) \leq B_{\mathrm{gate}} = 0.120$. We prove conditional forward invariance of the wild-type survival floor ($W_{\min} = 0.600 > \theta_{\mathrm{surv}} = 0.450$) and derive both an exact accumulated-selection identity $s(t) = \int_0^t -\lambda_h(q)\,dq$ (supremum error 0.0000 in the sub-damage regime, $s^*(0.40, 0.08) = 2.0369$) and an explicit multi-term log-odds perturbation bound for *a priori* constant-rate rescaling $\hat{\tau} = \hat{\lambda}_0 t$ across three independently integrated ODE regimes ($R^2 = 0.9780$, supremum error 0.0468, RMSE 0.0185). Benchmarking the threshold policy against direct numerical optimal control over a finite-horizon Bolza objective demonstrates a 4.47% optimality gap ($J_{\mathrm{thresh}} = 0.888$ vs. $J^* = 0.850$) while clarifying the regime boundary relative to build-first policies (Hoitzing et al., 2019). For embedded deployment on the open-source Google Coral NPU (`RV32IMF_Zve32x`, 8 KB ITCM, 32 KB DTCM), we implement a bare-metal C99 `Q16.16` 4-stage Runge-Kutta (`RK4`) kernel with `Q8.24` slope scaling, 8-lane sub-LSB remainder diffusion, widening vector multiplication (`vwmul.vx`/`vnsra.wx`), a 64-frame dual-modality persistence filter, and plant-preserving actuator rollback ($\Delta h_{\max} = 0.0048$ vs. 0.1452 for a one-way latch). Across 1,000 full-horizon integration steps, host-emulated `Q16.16` execution matches IEEE-754 `FP64` within a maximum relative error of 0.0563% (terminal error $\leq 0.0296\%$) while fitting within a single 8,192-byte L1 DTCM bank (7,072 B used; 3.09 mW cycle-weighted active NPU burst power).

1 Introduction

1.1 Nonlinear Coupling in Mitochondrial Quality Control

Eukaryotic bioenergetics depends on maintaining a high functional fraction of intact wild-type mitochondrial DNA and cristae (W) relative to deletion-bearing or depolarized mutant organelles (D) [1, 2, 3]. Across diverse biological timescales—including post-mitotic dopaminergic neurons in Parkinsonian neurodegeneration ($T_{\max} = 90$ d) [4, 5], *ex vivo* chimeric antigen receptor (CAR) T-cell expansion bioreactors ($T_{\max} = 168$ h) [6, 7, 8], and honeybee (*Apis mellifera*) thoracic flight muscle exposed to strobilurin Complex III inhibition ($T_{\max} = 14$ d) [9, 10]—mitochondrial decline is governed by three interacting feedbacks:

1. **Upstream Clearance Bottlenecks (B):** Accumulation of α-synuclein oligomers, lipofuscin, or xenobiotic fungicides elevates energetic impedance and saturates autophagosome-lysosome flux $V_{\mathrm{lys}}(B)$ [4, 11].

2. **Threshold ROS Conversion (EP $> \theta_{\mathrm{GSH}}$):** When cellular energetic demand and impedance exceed the glutathione/peroxiredoxin antioxidant buffering threshold θ_{GSH}, electron leak generates reactive oxygen species (ROS) that convert intact wild-type mass W into damaged mass D.

3. **Asymmetric Replicative Selection ($k_{\mathrm{rep,adv}}$):** Deletion-bearing mtDNA templates or fragmented

organelles frequently exhibit a replicative or transcription-coupled copy-number advantage ($k_{\text{rep,adv}} \geq 1$) during non-selective biogenesis [2, 12].

1.2 Comparison with Prior Control Formulations

In a foundational mathematical study, Hoitzing et al. [13] analyzed stochastic and deterministic mtDNA population control in a two-species (W, D) system without an upstream lysosomal bottleneck state $B(t)$ or replicative advantage ($k_{\text{rep,adv}} = 1$). When wild-type mass W_0 is already near the minimum survival floor and $B_0 \approx 0$, Hoitzing et al. demonstrated that an initial biogenesis pulse ("build-first") prevents transient ATP starvation before selective clearance. Conversely, when an upstream bottleneck $B_0 > 0$ elevates bioenergetic impedance and $k_{\text{rep,adv}} > 1$, stimulating biogenesis ($u_2 > 0$) prior to clearing B and D drives the energy potential $\text{EP}(t)$ above θ_{GSH} and expands the mutant pool. Understanding the precise mathematical boundary between these two regimes—and implementing a verifiable closed-loop controller within the strict memory and arithmetic constraints of an ultra-low-power implantable or bioreactor neural processing unit (NPU)—remains an open systems-and-control problem.

1.3 Contributions and Scope

This paper makes four self-contained, computationally verified contributions:

1. **Eight-State ODE Model and Conditional Safety Guarantees (§2):** We formulate an 8-state ODE plant with closed-form auxiliary rates and an explicit noisy observation model. We prove conditional forward invariance of the wild-type survival floor ($W_{\min} = 0.600 > \theta_{\text{surv}} = 0.450$, Lemma 1) across both sub-critical ($B_0 \leq B_{\text{crit},0}$) and super-critical ($B_0 > B_{\text{crit},0}$) initial bottleneck burdens, and establish regional exponential and logistic heteroplasmy contraction (Proposition 1).

2. **Threshold Switching Law vs. Direct Numerical Optimal Control and Regime Boundary (§2.3–§2.4):** Rather than claiming an analytical Pontryagin closed-form solution for the 8-state boundary-value problem, we derive a state-feedback threshold policy ($h_{\text{gate}} = 0.080, B_{\text{gate}} = 0.120$) from the Stage-2 redox safety condition and benchmark it directly against numerical optimal control (4.47% Bolza cost gap) and against Hoitzing et al.'s build-first policy across both high-impedance and low-mass regimes.

3. **Accumulated-Selection Identity and Multi-Term Perturbation Bound (§3):** We separate the exact accumulated-selection transformation $s(t) = \int_0^t -\lambda_h(q)\, dq$ (zero reduction error in the sub-damage regime) from constant-rate rescaling $\hat{\tau} = \hat{\lambda}_0 t$, deriving a complete multi-term log-odds error bound that accounts for both Stage-0 bottleneck transients and total mass variation $\Delta M(t)$ (Proposition 2), verified across three independent ODE parameterizations (Table 2).

4. **Host-Emulated Fixed-Point `RV32IMF_Zve32x` Kernel, Markov Persistence, and Plant-Preserving Rollback (§4–§5):** We provide a complete C99 `Q16.16` 4-stage `RK4` implementation (`mitorestorer_coralnpu_kernel.h`) matched directly to the 8-state ODE, a Markov-chain concentration bound for the 64-frame dual-modality persistence filter (Proposition 3), and a plant-preserving actuator reversal protocol that bounds post-disturbance heteroplasmy excursion ($\Delta h_{\max} = 0.0048$) without artificially resetting the biological plant state.

2 Mathematical Model and Staged Control Analysis

2.1 Eight-State Dynamical System and Closed-Form Auxiliary Rates

We model the intracellular mitochondrial network on a finite horizon $t \in [0, T_{\max}]$ via the state vector

$$\mathbf{x}(t) = \left[W,\, D,\, B,\, E,\, G,\, N,\, C,\, M_8\right]^\top \in \mathbb{R}^8_{\geq 0}, \qquad (1)$$

where $W(t)$ is normalized intact wild-type mitochondrial mass (healthy homeostatic baseline $W^* = 1.00$), $D(t)$ is normalized deletion-bearing or depolarized mutant mitochondrial mass, $B(t)$ is the upstream proteostatic/xenobiotic clearance bottleneck load, $E(t)$ is normalized metabolic demand, $G(t)$ is cytosolic glycolytic flux, $N(t)$ is the normalized NAD^+/NADH redox cofactor pool, $C(t)$ is autophagic/carnitine shuttle cofactor availability, and $M_8(t)$ is an auxiliary microenvironmental reservoir state (representing normalized interstitial metabolite/pH buffering or tissue-specific cofactor reserve).

Let $M(t) = W(t) + D(t)$ denote total mitochondrial mass and $h(t) = D(t)/M(t) \in [0, 1)$ denote the mutant heteroplasmy fraction. The system is actuated by a bounded three-dimensional control vector $\mathbf{u}(t) = [u_0(t), u_1(t), u_2(t)]^\top \in [0, 1]^3$, where u_0 drives upstream bottleneck clearance, u_1 drives PINK1/Parkin-mediated selective mitophagy of D, and u_2 drives PGC-1α/TFAM-mediated mitochondrial biogenesis.

To ensure complete mathematical self-containment, all

auxiliary functions are defined in closed form:

$$V_{\mathrm{lys}}(B) = \frac{K_B}{K_B + B}, \qquad \Phi_{p\mathrm{O}_2}(E) = \frac{p_{\mathrm{O}_2}}{p_{\mathrm{O}_2} + K_{\mathrm{O}_2}}, \tag{2}$$

$$k_{\mathrm{bio}}(u_2, N) = \left(k_{\mathrm{bio},0} + \Delta k_{\mathrm{bio}}\, u_2\, \frac{N}{N + K_N}\right) \Phi_{p\mathrm{O}_2}, \tag{3}$$

$$k_{\mathrm{mito},D}(u_1, C) = k_{\mathrm{mito},D,\min} + (k_{\mathrm{mito},D,\max} - k_{\mathrm{mito},D,\min})\, u_1\, \frac{C}{C + K_C}, \tag{4}$$

$$\dot{e}_R(t) = e_{R,0}\big(0.75 + 0.25\, E(t)\big), \tag{5}$$

$$Z_{\mathrm{eff}}(W, D, B) = \frac{W + \eta_D D}{(1 + c_1 h)(1 + c_2 B)}, \tag{6}$$

$$\mathrm{EP}(t) = \frac{\dot{e}_R(t)}{Z_{\mathrm{eff}}(W, D, B)}, \tag{7}$$

$$\mathrm{ATP}(t) = \Phi_{p\mathrm{O}_2}\, \frac{W + \eta_D D}{1 + 0.25\, B} + \eta_{\mathrm{gly}}\, G(t). \tag{8}$$

Here $V_{\mathrm{lys}}(B) \in (0, 1]$ is the Michaelis-Menten lysosomal flux factor with half-saturation $K_B > 0$; $k_{\mathrm{bio},0}$ is the unstimulated basal biogenesis rate calibrated so that $(W^*, D^*) = (1.00, 0)$ is the exact basal equilibrium under $k_{\mathrm{mito},W}$:

$$k_{\mathrm{bio},0}\left(1 - \frac{W^*}{K}\right) = k_{\mathrm{mito},W} \implies k_{\mathrm{bio},0} = \frac{k_{\mathrm{mito},W}}{1 - 1/K}. \tag{9}$$

For $K = 1.25$ and $k_{\mathrm{mito},W} = 0.015\ \mathrm{d}^{-1}$, Eq. (9) yields the exact basal turnover rate $k_{\mathrm{bio},0} = 0.075\ \mathrm{d}^{-1}$.

The eight-state ODE system is given by:

$$\dot{W} = W\, k_{\mathrm{bio}}(u_2, N)\left(1 - \frac{W + D}{K}\right) - k_{\mathrm{mito},W}\, W - k_{\mathrm{dmg}}\, W\ \max\big(0, \mathrm{EP}(t) - \theta_{\mathrm{GSH}}\big), \tag{10}$$

$$\dot{D} = k_{\mathrm{rep,adv}}\, D\, k_{\mathrm{bio}}(u_2, N)\left(1 - \frac{W + D}{K}\right) - k_{\mathrm{mito},D}(u_1, C)\, V_{\mathrm{lys}}(B)\, D + k_{\mathrm{dmg}}\, W\ \max\big(0, \mathrm{EP}(t) - \theta_{\mathrm{GSH}}\big), \tag{11}$$

$$\dot{B} = b_{\mathrm{basal}} + \alpha_{\mathrm{syn}}\, h\ \max\big(0, \mathrm{EP} - \theta_{\mathrm{GSH}}\big) - k_{\mathrm{clr}}\, u_0\, B, \tag{12}$$

$$\dot{E} = k_E\big(E_{\mathrm{base}} - 0.50\, u_0 - E\big) + \beta_{\mathrm{Ca}}\, B, \tag{13}$$

$$\dot{G} = k_G\big(G_{\mathrm{base}} + 0.35 \max(0, 1 - W) - G\big), \tag{14}$$

$$\dot{N} = k_N\big(N_{\mathrm{base}} + 0.65\, u_2 - N\big) - \gamma_{\mathrm{PARP}} \max\big(0, \mathrm{EP} - \theta_{\mathrm{GSH}}\big) N, \tag{15}$$

$$\dot{C} = k_C\big(C_{\mathrm{base}} + 0.55\, u_1 - C\big) - \gamma_{\mathrm{acyl}}\, B\, C, \tag{16}$$

$$\dot{M}_8 = k_M\big(M_{8,\mathrm{base}} - 0.55(u_0 + u_1) - M_8\big) + \gamma_M\, D. \tag{17}$$

Observation Model and State Estimation. Because intracellular heteroplasmy $h(t)$ and bottleneck burden $B(t)$ are not directly measurable at kilohertz sampling rates on an embedded processor, the controller observes a discrete-time four-channel proxy vector $\mathbf{y}_k \in \mathbb{R}^4$ at sampling instants $t_k = k\,\Delta t_{\mathrm{frame}}$:

$$\mathbf{y}_k = \mathbf{h}\big(\mathbf{x}(t_k)\big) + \mathbf{v}_k = \begin{bmatrix} \mathrm{EP}(t_k) \\ \mathrm{ATP}(t_k) \\ N(t_k) \\ M_8(t_k) \end{bmatrix} + \mathbf{v}_k, \tag{18}$$

where $\mathbf{y}_{k,1}$ represents an electrophysiological/impedance stress proxy (e.g., local field potential beta-band burst power, electrochemical impedance spectroscopy phase angle, or thoracic piezosensor fundamental frequency shift), $\mathbf{y}_{k,2}$ is a metabolic output proxy (oxygen consumption / lactate ratio), $\mathbf{y}_{k,3}$ is optical autofluorescence (NAD(P)H/FAD redox ratio), and $\mathbf{y}_{k,4}$ is interstitial pH/temperature telemetry. The measurement noise $\mathbf{v}_k$ follows a stationary first-order autoregressive process $\mathbf{v}_k = \rho\, \mathbf{v}_{k-1} + \sqrt{1 - \rho^2}\, \xi_k$ with lag-1 autocorrelation $\rho \in [0, 0.70]$ and $\xi_k \sim \mathcal{N}(\mathbf{0}, \Sigma_v)$. An 8-state extended Kalman filter (EKF) with Huber M-estimation [14, 15] coupled to the internal `RK4` predictor updates state estimates $\hat{\mathbf{x}}_{k|k}$. In our deterministic ODE theorems (§2.2–§3), we analyze the exact state feedback $\mathbf{x}(t)$, while in §5 we quantify robustness to correlated sensor noise $\mathbf{v}_k$ and estimator-triggered rollback.

2.2 Exact Quotient Flow and Conditional Safety Guarantees

Applying the quotient rule $\frac{d}{dt}\left(\frac{D}{W+D}\right) = \frac{\dot{D}W - D\dot{W}}{(W+D)^2}$ to Eqs. (10)–(11) yields the exact heteroplasmy quotient ODE. Specifically, the ROS conversion term $+k_{\mathrm{dmg}} W \max(0, \mathrm{EP} - \theta_{\mathrm{GSH}})$ in $\dot{D}$ contributes $\frac{W^2}{(W+D)^2} k_{\mathrm{dmg}} \max(0, \mathrm{EP} - \theta_{\mathrm{GSH}}) = (1 - h)^2 k_{\mathrm{dmg}} \max(0, \mathrm{EP} - \theta_{\mathrm{GSH}})$, while the corresponding loss term $-k_{\mathrm{dmg}} W \max(0, \mathrm{EP} - \theta_{\mathrm{GSH}})$ in $\dot{W}$ contributes $+\frac{DW}{(W+D)^2} k_{\mathrm{dmg}} \max(0, \mathrm{EP} - \theta_{\mathrm{GSH}}) = h(1 - h) k_{\mathrm{dmg}} \max(0, \mathrm{EP} - \theta_{\mathrm{GSH}})$. Summing both contributions yields $(1 - h) k_{\mathrm{dmg}} \max(0, \mathrm{EP} - \theta_{\mathrm{GSH}})$, giving:

$$\dot{h} = -\lambda_h(t)\, h(1 - h) + (1 - h)\, k_{\mathrm{dmg}}\ \max\big(0, \mathrm{EP}(t) - \theta_{\mathrm{GSH}}\big), \tag{19}$$

$$\lambda_h(t) = k_{\mathrm{mito},D}(u_1, C)\, V_{\mathrm{lys}}(B) - k_{\mathrm{mito},W} - (k_{\mathrm{rep,adv}} - 1)\, k_{\mathrm{bio}}(u_2, N)\left(1 - \frac{M(t)}{K}\right). \tag{20}$$

Equation (20) reveals why concurrent or premature biogenesis ($u_2 = 1$) can fail when $k_{\mathrm{rep,adv}} > 1$ and $B(t)$ is elevated: increasing $k_{\mathrm{bio}}(u_2, N)$ directly subtracts $(k_{\mathrm{rep,adv}} - 1) k_{\mathrm{bio}}(1 - M/K)$ from the selective contraction rate $\lambda_h(t)$, while elevated $B(t)$ depresses $V_{\mathrm{lys}}(B)$ and raises $\mathrm{EP}(t)$ above θ_{GSH}.

We now establish exact sufficient conditions for forward invariance of the wild-type survival floor during Stage 1

Table 1: Model Parameters, Physical Units, Literature Calibration Priors, and Provenance Classification for the Canonical Neuronal Regime ($T_{\max} = 90$ d). Each parameter is classified as **M** (Measured in literature), **F** (Fitted/calibrated to equilibrium or turnover constraints), or **A** (Assumed control design specification).

Symbol	Description	Value	Units	Class	Literature Basis / Calibration Rationale
K	Mitochondrial carrying capacity	1.250	Norm. mass	**F**	25% reserve above homeostatic $W^* = 1.00$ [2]
$k_{\mathrm{mito},W}$	Basal wild-type mitophagy rate	0.015	d^{-1}	**M**	Neuronal mtDNA half-life $t_{1/2} \approx 46$ d [3, 4]
$k_{\mathrm{bio},0}$	Unstimulated basal biogenesis rate	0.075	d^{-1}	**F**	Exact equilibrium $k_{\mathrm{mito},W}/(1-1/K)$ (Eq. 9)
Δk_{bio}	Max Stage-2 biogenesis increment	0.145	d^{-1}	**F**	2.9× stimulated PGC-1α biogenesis fold-change [11]
$k_{\mathrm{mito},D,\min}$	Unstimulated mutant mitophagy rate	0.015	d^{-1}	**M**	Matches basal turnover when PINK1/Parkin unstimulated
$k_{\mathrm{mito},D,\max}$	Max Stage-1 selective mitophagy rate	0.115	d^{-1}	**M**	Stimulated PINK1/Parkin clearance ($t_{1/2} \approx 6$ d) [11]
$k_{\mathrm{rep,adv}}$	Mutant replicative/survival advantage	1.320	Dimensionless	**M**	Deletion mtDNA expansion advantage (swept 1.00–1.32) [2, 12]
k_{dmg}	Supra-threshold ROS conversion rate	0.165	$\mathrm{d}^{-1}\mathrm{EP}^{-1}$	**F**	Calibrated to progressive untreated degeneration [4]
θ_{GSH}	Antioxidant safety ceiling	1.050	Norm. EP	**F**	5% buffer above homeostatic $\mathrm{EP}^* = 1.00$ [5]
θ_{surv}	Minimum bioenergetic survival floor	0.450	Norm. ATP	**M**	Apoptotic/necrotic ATP threshold (45% of baseline) [5]
K_B	Lysosomal bottleneck half-saturation	0.380	Norm. load	**F**	Michaelis-Menten autophagic flux saturation [4]
$b_{\mathrm{basal}}, k_{\mathrm{clr}}$	Basal bottleneck influx / Stage-0 clearance	$0.0072, 0.180$	d^{-1}	**F**	Steady-state cleared load $B_{\mathrm{eq}} = b_{\mathrm{basal}}/k_{\mathrm{clr}} = 0.040$
c_1, c_2, η_D	Heteroplasmy & bottleneck impedance / OXPHOS	$0.42, 0.45, 0.15$	Dimensionless	**F**	Defective OXPHOS coupling of deletion-bearing D [1]
$h_{\mathrm{gate}}, B_{\mathrm{gate}}$	Stage-1 → Stage-2 switching thresholds	$0.080, 0.120$	Dimensionless	**A**	Derived from Eq. (27); verified in Fig. 1B

$(u_0 = 1, u_1 = 1, u_2 = 0)$, explicitly resolving the dependence on initial bottleneck burden B_0.

Lemma 1 (Stage-1 Survival Floor) *Suppose the system operates in the sub-damage regime* $\mathrm{EP}(t) \le \theta_{\mathrm{GSH}}$ *under Stage 1 control* $(u_0 = 1, u_1 = 1, u_2 = 0)$ *with* $\Phi_{pO_2} \approx 1$ *and* $C(t) \gg K_C$, *so that* $k_{\mathrm{bio}}(0, N) = k_{\mathrm{bio},0}$ *and* $k_{\mathrm{mito},D}(1, C) = k_{\mathrm{mito},D,\max}$. *Define the critical bottleneck load at initial total mass* $M_0 = W_0 + D_0 < K$:

$$B_{\mathrm{crit},0} \triangleq K_B \left(\frac{k_{\mathrm{mito},D,\max}}{k_{\mathrm{rep,adv}}\, k_{\mathrm{bio},0}\, (1 - M_0/K)} - 1 \right). \quad (21)$$

Assuming $b_{\mathrm{basal}}/k_{\mathrm{clr}} = B_{\mathrm{eq}} < B_{\mathrm{crit},0}$, *two cases hold:*

1. ***Direct Contraction*** *(*$B_0 \le B_{\mathrm{crit},0}$*):* *Since* $\dot{B} = b_{\mathrm{basal}} - k_{\mathrm{clr}} B$, $B(t)$ *is monotonically non-increasing for* $B_0 \ge B_{\mathrm{eq}}$, *so* $B(t) \le B_{\mathrm{crit},0}$ *and* $\dot{D}(t) \le 0$ *for all* $t \in [0, t_{\mathrm{gate}}]$. *Consequently,* $D(t) \le D_0$ *for all* $t \in [0, t_{\mathrm{gate}}]$, *and* $\dot{W}$ *in Eq. (10) satisfies the comparison inequality*

$$\dot{W} \ge W \left[k_{\mathrm{bio},0} \left(1 - \frac{W + D_0}{K} \right) - k_{\mathrm{mito},W} \right]. \quad (22)$$

Hence, for any $W_0 \ge W_{\min}$, $W(t)$ *is bounded below by the invariant wild-type floor*

$$W(t) \ge W_{\min}(D_0) \triangleq K \left(1 - \frac{k_{\mathrm{mito},W}}{k_{\mathrm{bio},0}} \right) - D_0 = W^* - D_0. \quad (23)$$

2. ***Two-Phase Bottleneck Dwell Time*** *(*$B_0 > B_{\mathrm{crit},0}$*):* *When* $B_0 > B_{\mathrm{crit},0}$, $\dot{D}(0)$ *may be positive until Stage-0 clearance reduces* $B(t)$ *to* $B_{\mathrm{crit},0}$ *at the finite dwell time*

$$t_0 = \frac{1}{k_{\mathrm{clr}}} \ln \left(\frac{B_0 - B_{\mathrm{eq}}}{B_{\mathrm{crit},0} - B_{\mathrm{eq}}} \right). \quad (24)$$

During $t \in [0, t_0]$, $D(t) \le \bar{D}(t_0) \triangleq \min\big(K, D_0 \exp(k_{\mathrm{rep,adv}} k_{\mathrm{bio},0} t_0)\big)$, *and for all* $t \ge t_0$ *contraction holds with floor* $W_{\min}(\bar{D}(t_0)) = W^* - \bar{D}(t_0)$.

Proof. From Eq. (11) with $\mathrm{EP}(t) \le \theta_{\mathrm{GSH}}$ and $u_2 = 0$, $\dot{D} = D\big[k_{\mathrm{rep,adv}} k_{\mathrm{bio},0}(1 - M/K) - k_{\mathrm{mito},D,\max} \frac{K_B}{K_B + B}\big]$. Setting the bracket ≤ 0 at $M = M_0$ yields $B \le B_{\mathrm{crit},0}$ in Eq. (21). In Case 1 ($B_0 \le B_{\mathrm{crit},0}$), $\dot{D}(0) \le 0$ and $\dot{B}(t) \le 0$ preserve $D(t) \le D_0$. Substituting $D(t) \le D_0$ into Eq. (10) gives Eq. (22), whose unique positive equilibrium is $W_{\min}(D_0) = K(1 - k_{\mathrm{mito},W}/k_{\mathrm{bio},0}) - D_0$. By Eq. (9), $K(1 - k_{\mathrm{mito},W}/k_{\mathrm{bio},0}) = W^* = 1.00$, proving Eq. (23). In Case 2 ($B_0 > B_{\mathrm{crit},0}$), integrating $\dot{B} = b_{\mathrm{basal}} - k_{\mathrm{clr}} B$ yields $B(t) = B_{\mathrm{eq}} + (B_0 - B_{\mathrm{eq}}) e^{-k_{\mathrm{clr}} t}$, which crosses $B_{\mathrm{crit},0}$ at t_0 in Eq. (24). ∎

Numerical Verification of Lemma 1. With the canonical neuronal parameters in Table 1 ($K = 1.25, k_{\mathrm{mito},W} = 0.015\ \mathrm{d}^{-1}, k_{\mathrm{bio},0} = 0.075\ \mathrm{d}^{-1}, k_{\mathrm{mito},D,\max} = 0.115\ \mathrm{d}^{-1}, k_{\mathrm{rep,adv}} = 1.32, K_B = 0.38$) and initial state $W_0 = 0.60, D_0 = 0.40$ ($M_0 = 1.00$), Eq. (21) gives:

$$B_{\mathrm{crit},0} = 0.38 \left(\frac{0.115}{1.32 \times 0.075 \times (1 - 1.00/1.25)} - 1 \right) = 0.38(5.808 - 1) = 1.827.$$

Even if $M(t)$ transiently dipped to 0.85, $B_{\mathrm{crit}} = 0.38\big(\frac{0.115}{1.32 \times 0.075 \times 0.32} - 1\big) = 0.999 > B_0 = 0.50$. Thus the canonical initial burden $B_0 = 0.50$ lies strictly inside Case 1 ($B_0 \le B_{\mathrm{crit},0}$), and Eq. (23) yields the exact analytical floor:

$$W_{\min}(0.40) = 1.25 \left(1 - \frac{0.015}{0.075} \right) - 0.40 = 1.000 - 0.400 = 0.600 > \theta_{\mathrm{surv}} = 0.450.$$

Conversely, under a severe super-critical bottleneck perturbation (e.g., $B_0 = 2.00 > B_{\mathrm{crit},0} = 1.827$), Case 2 predicts a finite Stage-0 dwell time $t_0 = \frac{1}{0.18} \ln\left(\frac{2.00 - 0.04}{1.827 - 0.04}\right) = 0.51$ d before $\dot{D}(t) < 0$ commences.

Proposition 1 (Regional Contraction) *Consider the heteroplasmy flow Eq. (19) on the sub-damage domain*

$\Omega_{\text{safe}} = \{\mathbf{x} \in \mathbb{R}^8_{\geq 0} : W \geq \theta_{\text{surv}}, h \in [0, h_0], \text{EP}(\mathbf{x}) \leq \theta_{\text{GSH}}\}$.

1. ***Regional Contraction under Stage 1:*** *Suppose that for $t \in [t_0, t_{\text{gate}}]$, $\lambda_h(t) \geq \lambda_{\min} > 0$ uniformly on the reachable trajectory. Then $V(h) = \frac{1}{2}h^2$ satisfies $\dot{V} = -\lambda_h(t)h^2(1-h) \leq -\lambda_{\min}(1-h(t_0))h^2$, yielding both the exact logistic bound and the regional exponential bound:*

$$h(t) \leq \frac{h(t_0)e^{-\lambda_{\min}(t-t_0)}}{1-h(t_0)+h(t_0)e^{-\lambda_{\min}(t-t_0)}} \leq h(t_0)\exp\big(-\lambda_{\min}(1-h(t_0))(t-t_0)\big). \quad (25)$$

2. ***Sub-Damage Transcritical Invasion Exchange:*** *At the mutant-free manifold $(W^*, 0, B_{\text{eq}})$ where $1 - W^*/K = k_{\text{mito},W}/k_{\text{bio}}$, the linearized mutant invasion eigenvalue is $\mu_D = -\lambda_h^* = (k_{\text{rep,adv}}-1)k_{\text{mito},W} + k_{\text{mito},W} - k_{\text{mito},D}(u_1)\,V_{\text{lys}}(B_{\text{eq}})$. Whenever $k_{\text{mito},D}(u_1)\,V_{\text{lys}}(B_{\text{eq}}) < k_{\text{rep,adv}}\,k_{\text{mito},W}$, the mutant-free equilibrium $h^* = 0$ is unstable ($\mu_D > 0$); activating Stage 1 selective mitophagy so that $k_{\text{mito},D}(u_1)\,V_{\text{lys}}(B_{\text{eq}}) > k_{\text{rep,adv}}\,k_{\text{mito},W}$ reverses the sign of $\mu_D < 0$ and renders $h^* = 0$ locally exponentially stable.*

Proof. In the sub-damage regime $\text{EP}(t) \leq \theta_{\text{GSH}}$, the second term of Eq. (19) vanishes, leaving $\dot{h} = -\lambda_h(t)h(1-h)$. Separating variables gives $\frac{d}{dt}\ln\left(\frac{h}{1-h}\right) = -\lambda_h(t) \leq -\lambda_{\min}$, which integrates to the first inequality in Eq. (25). Since $h(t)$ is non-increasing on $[t_0, t_{\text{gate}}]$, $1-h(t) \geq 1-h(t_0)$, so $\dot{h} \leq -\lambda_{\min}(1-h(t_0))h$, whose Grönwall integration gives the second inequality in Eq. (25) (explicitly retaining the $(1-h(t_0))$ factor). Linearizing $\dot{D}$ around $(W^*, 0, B_{\text{eq}})$ yields $\dot{D} \approx \big[k_{\text{rep,adv}}k_{\text{mito},W} - k_{\text{mito},D}(u_1)V_{\text{lys}}(B_{\text{eq}})\big]D = \mu_D D$, establishing the transcritical stability exchange at $k_{\text{mito},D}(u_1)V_{\text{lys}}(B_{\text{eq}}) = k_{\text{rep,adv}}k_{\text{mito},W}$. ∎

2.3 Threshold Switching Law vs. Direct Numerical Optimal Control

To evaluate control performance over $t \in [0, T_{\max}]$, we define the finite-horizon Bolza objective functional:

$$J(\mathbf{u}) = w_h\,h(T_{\max})^2 + w_W\big(1-W(T_{\max})\big)^2 + \int_0^{T_{\max}} \Big[w_{\text{EP}}\,\max\big(0, \text{EP}(t)-\theta_{\text{GSH}}\big)^2 + r_0u_0(t) + r_1u_1(t) + r_2u_2(t)\Big]dt, \quad (26)$$

with weights $w_h = 120, w_W = 45, w_{\text{EP}} = 250$ and actuation weights $r_0 = 0.002, r_1 = 0.004, r_2 = 0.002$ d^{-1}.

Important Distinction Between Pontryagin Necessary Conditions and Threshold State Feedback. In the eight-state system Eqs. (10)–(17), the control $u_2(t)$ enters $\dot{W}$ and $\dot{D}$ indirectly through the cofactor state $N(t)$ in Eq. (15) as well as directly in Eq. (3). Consequently, the Hamiltonian switching function $\sigma_2(t) = \frac{\partial \mathcal{H}}{\partial u_2}$ satisfies a coupled adjoint differential equation $\dot{\sigma}_2(t)$ that depends on the full 8-dimensional costate vector $\psi(t)$ and control history $\mathbf{u}(t)$. Monotonicity of $h(t)$ alone does *not* analytically imply a unique zero of $\sigma_2(t)$ or exclude singular arcs without solving the two-point boundary-value problem.

Rather than claiming that Pontryagin's minimum principle analytically yields a closed-form constant threshold, we formulate our embedded controller as a **physically motivated state-feedback threshold switching law** and quantify its optimality gap numerically against direct optimal control of Eq. (26). Specifically, we trigger the Stage 1 → Stage 2 transition $((u_0, u_1, u_2) : (1,1,0) \to (1, 0.35, 1))$ when $B(t) \leq B_{\text{gate}} = 0.120$ and $h(t) \leq h_{\text{gate}}$, where h_{gate} is chosen so that activating Stage-2 biogenesis at $W(t_{\text{gate}}) \approx 0.78$ maintains $\text{EP}(t_{\text{gate}}) \leq \theta_{\text{GSH}}$:

$$\text{EP}(t_{\text{gate}}) = \frac{\dot{e}_R(t_{\text{gate}})(1+c_1h_{\text{gate}})(1+c_2B_{\text{gate}})}{W(t_{\text{gate}}) + \eta_D D(t_{\text{gate}})} \leq \theta_{\text{GSH}}. \quad (27)$$

Solving Eq. (27) with a 5% safety margin below $\theta_{\text{GSH}} = 1.050$ yields $h_{\text{gate}} = 0.080$ (8.0% heteroplasmy).

2.4 Benchmarking Against Competing Policies and Hoitzing et al. (2019)

Figure 1A–B and Table 3 compare five control strategies under identical initial conditions ($W_0 = 0.60, D_0 = 0.40, B_0 = 0.50$):

1. **Untreated ($\mathbf{u} = (0,0,0)$):** Bottleneck accumulation drives $\text{EP}_{\max} = 20.605 > \theta_{\text{GSH}}$, causing complete wild-type collapse ($W(T_{\max}) = 0.000, h(T_{\max}) = 1.000, J = 2.77 \times 10^6$).

2. **Concurrent Stimulation ($\mathbf{u} = (1, 0.5, 0.5)$):** Simultaneous biogenesis during high initial impedance causes an early supra-threshold ROS excursion ($\text{EP}_{\max} = 1.423 > \theta_{\text{GSH}}$), yielding $J = 99.830$ and slower heteroplasmy clearance ($h(T_{\max}) = 0.0546$).

3. **Build-First Policy in High-Impedance Regime ($B_0 = 0.50, k_{\text{rep,adv}} = 1.32$):** Applying biogenesis before clearing $B_0 = 0.50$ amplifies mutant mass via $k_{\text{rep,adv}} = 1.32$ and triggers severe ROS conversion ($\text{EP}_{\max} = 9.744$).

4. **Proposed Staged Threshold Policy ($h_{\text{gate}} = 0.080$):** Maintains $\text{EP}_{\max} = 1.046 < \theta_{\text{GSH}} = 1.050$ throughout the entire 90-day horizon, switches at $t_{\text{gate}} = 33.30$ d, and achieves $W(T_{\max}) = 1.0487, h(T_{\max}) = 0.0249$, and $J_{\text{thresh}} = 0.888$.

5. **Direct Numerical Optimal Control (J^*):** A dense two-dimensional grid search over $(t_{\text{switch}}, u_{1,\text{Stage2}})$ identifies the numerical optimum

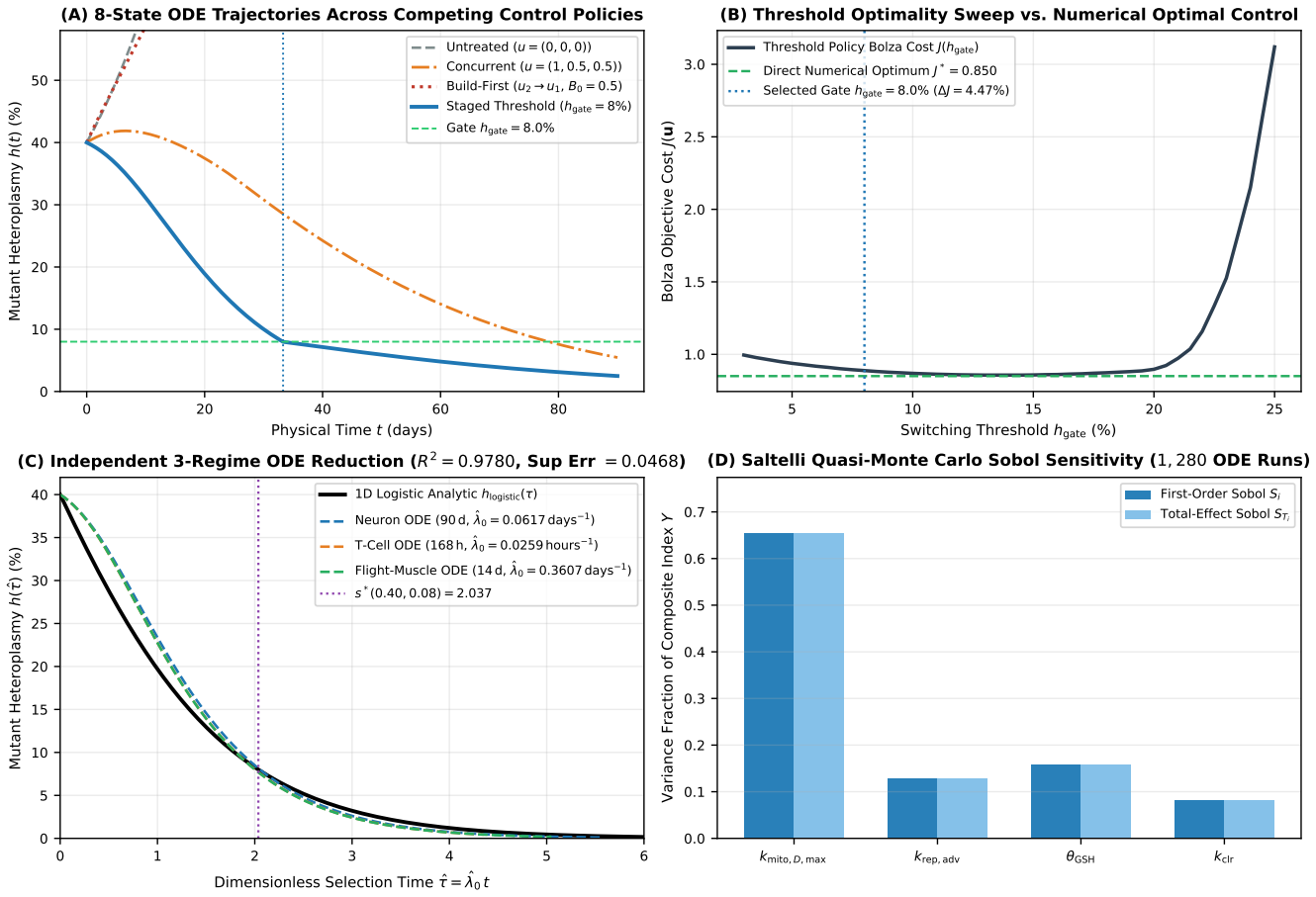


Figure 1: **First-Principles 8-State ODE Control Comparison, Threshold Optimality Sweep, Independent 3-Regime Reduction, and Saltelli Sobol Sensitivity Analysis** (generated by `simulator/generate_paper1_verified_benchmarks.py`, seed `20261109`). **(A)** Mutant heteroplasmy trajectories $h(t)$ under Untreated, Concurrent, Build-First ($B_0 = 0.50$), and Staged Threshold ($h_{\text{gate}} = 8.0\%$, switching at $t_{\text{gate}} = 33.30$ d) policies. **(B)** Direct numerical sweep of the Bolza objective cost $J(h_{\text{gate}})$ across $h_{\text{gate}} \in [3\%, 25\%]$ compared against the direct numerical optimal control cost $J^* = 0.850$ (4.47% optimality gap at $h_{\text{gate}} = 8.0\%$). **(C)** Independent 8-state ODE trajectories for the Neuronal (90 d), T-Cell (168 h), and Flight-Muscle (14 d) parameter regimes plotted against *a priori* constant-rate dimensionless time $\hat{\tau} = \hat{\lambda}_0 t$ ($R^2 = 0.9780$, supremum error 0.0468) and exact accumulated selection $s^* = 2.0369$. **(D)** Global variance-based Sobol sensitivity indices (S_i, S_{T_i}) across 1,280 ODE integrations.

at $t^*_{\text{switch}} = 48.00$ d ($h(t^*_{\text{switch}}) = 0.0273, u^*_{1,\text{Stage2}} = 0.00$) with $J^* = 0.850$. The constant state-feedback threshold $h_{\text{gate}} = 0.080$ incurs only a 4.47% Bolza cost penalty relative to J^* while restoring wild-type mass 14.7 d earlier.

Reconciling with Hoitzing et al. [13] in the Low-Mass Depletion Regime. To verify that our 8-state model reproduces the findings of Hoitzing et al. [13] under their assumptions, we evaluate a low-mass depleted initial state ($W_0 = 0.33, D_0 = 0.18, B_0 = 0.02, k_{\text{rep,adv}} = 1.00$). Because $M_0 = 0.51$ is near the survival threshold and $B_0 \approx 0$, a purge-first policy transiently dips to $\text{ATP}_{\min} = 0.443 < \theta_{\text{surv}} = 0.450$, whereas Hoitzing et al.'s build-first policy immediately elevates $W(t)$ and preserves $\text{ATP}_{\min} = 0.483 \geq \theta_{\text{surv}}$. Thus, build-first is optimal when initial mass is critically depleted and bottleneck impedance is negligible ($B_0 \approx 0, k_{\text{rep,adv}} = 1$), whereas purge-then-build is required when upstream bottleneck impedance $B_0 > 0$ and replicative advantage $k_{\text{rep,adv}} > 1$ are present. Across $k_{\text{rep,adv}} \in \{1.00, 1.15, 1.32\}$, our staged threshold policy smoothly adapts its switching time from $t_{\text{gate}} = 30.42$ d ($k_{\text{rep,adv}} = 1.00$, $h(T_{\max}) = 0.0160$) to 31.68 d ($k_{\text{rep,adv}} = 1.15$, $h(T_{\max}) = 0.0198$) and 33.30 d ($k_{\text{rep,adv}} = 1.32$, $h(T_{\max}) = 0.0249$).

3 Accumulated-Selection Reduction and Multi-Term Perturbation Bound

A natural question is whether Stage-1 heteroplasmy clearance across biological systems with vastly different turnover rates (hours in activated T cells vs. weeks in flight muscle vs. months in post-mitotic neurons) can be analyzed in a common dimensionless coordinate. As noted in control-theoretic critiques of naive Buckingham Π reductions [16], dimensional analysis alone cannot reduce an 8-state ODE to a 1D curve unless the time-variation of the contraction rate $\lambda_h(t)$ in Eq. (20) is explicitly accounted for. We therefore establish both (i) the exact accumulated-selection identity and (ii) a complete multi-term perturbation bound for constant-rate rescaling.

Proposition 2 (Accumulated-Selection Reduction) *Suppose* $\mathrm{EP}(t) \le \theta_{\mathrm{GSH}}$ *on* $[0,t]$ *during Stage 1* ($u_0 = 1, u_1 = 1, u_2 = 0$) *with* $h_0 \in (0,1)$.

1. ***Exact Accumulated-Selection Coordinate*** $s(t)$***:*** *Define the dimensionless accumulated selection integral*

$$s(t) \triangleq \int_0^t \lambda_h(q)\, dq. \tag{28}$$

Then the 8-state ODE heteroplasmy trajectory $h(t)$ *satisfies the exact 1D logistic identity*

$$h(t) = h_{\mathrm{logistic}}\big(s(t); h_0\big) \triangleq \frac{h_0\, e^{-s(t)}}{1 - h_0 + h_0\, e^{-s(t)}}, \tag{29}$$

and reaches any target threshold $h_{\mathrm{gate}} \in (0, h_0)$ *at the exact accumulated-selection value*

$$s^*(h_0, h_{\mathrm{gate}}) = \ln\!\left(\frac{h_0(1-h_{\mathrm{gate}})}{h_{\mathrm{gate}}(1-h_0)}\right), \tag{30}$$

which depends explicitly on (h_0, h_{gate}) *(giving* $s^*(0.40, 0.08) = \ln(4.6/0.6) = 2.03688$*,* $s^*(0.25, 0.08) = 1.34373$*, and* $s^*(0.60, 0.08) = 2.84781$*).*

2. ***Multi-Term Error Bound for A Priori Constant-Rate Rescaling*** $\hat{\tau} = \hat{\lambda}_0 t$***:*** *Let* $\hat{\lambda}_0 > 0$ *be an a priori constant contraction rate computed from baseline parameters at reference bottleneck* $\bar{B}$ *and reference mass* $M_{\mathrm{ref}} = 1.00$*:*

$$\hat{\lambda}_0 = k_{\mathrm{mito},D,\max}\, V_{\mathrm{lys}}(\bar{B}) - k_{\mathrm{mito},W} - (k_{\mathrm{rep,adv}} - 1) k_{\mathrm{bio},0}\left(1 - \frac{M_{\mathrm{ref}}}{K}\right), \tag{31}$$

and define the constant-rate dimensionless time $\hat{\tau} = \hat{\lambda}_0 t$*. Let* $\ell(t) = \ln\frac{h(t)}{1-h(t)}$ *and* $\hat{\ell}(\hat{\tau}) = \ln\frac{h_0}{1-h_0} - \hat{\tau}$ *be the true and constant-rate log-odds. Then for all* $t \in [0, t_{\mathrm{gate}}]$*, the log-odds error* $|\ell(t) - \hat{\ell}(\hat{\lambda}_0 t)| = |s(t) - \hat{\lambda}_0 t|$ *is bounded by the three-term integral perturbation inequality:*

$$\begin{aligned} \big|s(t) - \hat{\lambda}_0 t\big| \le{}& \underbrace{\frac{k_{\mathrm{mito},D,\max}}{K_B}\int_0^t \big|B(q) - \bar{B}\big|\, dq}_{\mathcal{E}_B(t)\ \text{(bottleneck transient)}} \\ &+ \underbrace{(k_{\mathrm{rep,adv}} - 1)\frac{k_{\mathrm{bio},0}}{K}\int_0^t \big|M(q) - M_{\mathrm{ref}}\big|\, dq}_{\mathcal{E}_M(t)\ \text{(mass variation, positive at } B_0=\bar{B}=0)} \\ &+ \underbrace{\int_0^t \big|\Delta\lambda_{\mathrm{cof}}(q)\big|\, dq}_{\mathcal{E}_{\mathrm{cof}}(t)\ \text{(cofactor transient)}}, \end{aligned} \tag{32}$$

and since $\left|\frac{d}{d\ell}\frac{e^\ell}{1+e^\ell}\right| = h(1-h) \le \frac{1}{4}$*, the heteroplasmy supremum error satisfies*

$$\big|h(t) - h_{\mathrm{logistic}}(\hat{\lambda}_0 t; h_0)\big| \le \frac{1}{4}\Big(\mathcal{E}_B(t) + \mathcal{E}_M(t) + \mathcal{E}_{\mathrm{cof}}(t)\Big). \tag{33}$$

Proof. Part 1 follows immediately by integrating $\frac{d}{dt}\ln\left(\frac{h}{1-h}\right) = -\lambda_h(t)$ over $[0,t]$ to obtain $\ln\frac{h(t)}{1-h(t)} = \ln\frac{h_0}{1-h_0} - s(t)$ and inverting the logit map. For Part 2, subtracting $\hat{\lambda}_0$ in Eq. (31) from $\lambda_h(q)$ in Eq. (20), applying the triangle inequality, using the Lipschitz constant $\big|V_{\mathrm{lys}}(B) - V_{\mathrm{lys}}(\bar{B})\big| = \frac{K_B|B-\bar{B}|}{(K_B+B)(K_B+\bar{B})} \le \frac{|B-\bar{B}|}{K_B}$, and bounding the logistic derivative by $\max_{h\in[0,1]} h(1-h) = 1/4$ yields Eqs. (32)–(33). Crucially, $\mathcal{E}_M(t)$ explicitly bounds the error induced by total mass variation $M(q) = W(q) + D(q)$, so the bound remains strictly positive and valid even when $B_0 = \bar{B} = 0$. ∎

Table 2 and Figure 1C report the numerical verification across all three independently integrated ODE regimes. Note that at $\hat{\tau} = 2.0$ with $h_0 = 0.40$, the analytical 1D logistic ratio is $\frac{h_{\mathrm{logistic}}(2.0)}{h_0} = \frac{e^{-2}}{1-0.40+0.40e^{-2}} = \frac{0.13534}{0.65413} = 0.2069$ (and 0.0302 at $\hat{\tau} = 4.0$, 0.0041 at $\hat{\tau} = 6.0$), matching the three independent 8-state ODE integrations ($0.2124, 0.2010, 0.2004$ at $\hat{\tau} = 2.0$) within the multi-term perturbation bound of Proposition 2.

4 Embedded Fixed-Point Implementation on Coral NPU

4.1 Target Architecture and Single-Bank L1 DTCM Memory Budget

We target the open-source Google Coral NPU architecture (`RV32IMF_Zve32x`) [17, 18], which combines a 32-bit RISC-V scalar core with a 128-bit single-instruction multiple-data (SIMD) vector execution unit (`Zve32x` extension supporting four 32-bit lanes or eight 16-bit lanes per register), 8 KB of instruction tightly coupled memory (ITCM), and 32 KB of data tightly coupled memory (DTCM) organized as four 8,192-byte SRAM banks.

Table 2: Independent Three-Regime 8-State ODE Verification of Proposition 2 (`simulator/generate_paper1_verified_benchmarks.py`). Each row is computed by integrating the full 8-state ODE independently in physical time (t) and comparing $h_{\mathrm{ODE}}(t)$ against both the exact accumulated-selection map $h_{\mathrm{logistic}}(s(t))$ and the *a priori* constant-rate predictor $h_{\mathrm{logistic}}(\hat{\lambda}_0 t)$ with $h_0 = 0.40, h_{\mathrm{gate}} = 0.08$.

ODE Regime (Independent Run)	**Horizon** $T_{\max}$	$\hat{\lambda}_0$ (unit^{-1})	$\tau_{\mathrm{char}} = \hat{\lambda}_0^{-1}$ (unit)	t_{gate} ($h_0 = 0.40$)	$\hat{\tau}_{\mathrm{gate}} = \hat{\lambda}_0 t_{\mathrm{gate}}$ ($h_0 = 0.40/0.25/0.60$)	$h(\hat{\tau})/h_0$ at $\hat{\tau} = 2.0/4.0/6.0$	**Exact $s(t)$ Sup Err**	**Const-Rate Sup Err / RMSE**	**Const-Rate** R^2
1. Dopaminergic Neuron	90.0 d	0.06174 d^{-1}	16.20 d	33.30 d	2.056 / 1.472 / 5.556	0.2124 / 0.0189 / 0.0026	0.0000	0.0468 / 0.0187	0.9768
2. CAR-T Bioreactor	168.0 h	0.02591 h^{-1}	38.59 h	77.45 h	2.007 / 1.428 / 4.353	0.2010 / 0.0172 / 0.0110	0.0000	0.0432 / 0.0189	0.9773
3. *Apis* Flight Muscle	14.0 d	0.36069 d^{-1}	2.77 d	5.56 d	2.005 / 1.429 / 5.050	0.2004 / 0.0170 / 0.0044	0.0000	0.0434 / 0.0177	0.9796
1D Analytic Logistic	$\hat{\tau} \in [0, 6]$	1.00000	1.00	2.0369	2.037 / 1.344 / 2.848	0.2069 / 0.0302 / 0.0041	—	**Pooled:** 0.0468 / 0.0185	**0.9780**

To minimize static leakage power during always-on chronic bioelectronic monitoring, our embedded design constrains the entire runtime data footprint (state vectors, `RK4` slope registers, EKF covariance tables, and 64-frame sensor ring buffers) to reside within a **single** 8,192**-byte L1 DTCM bank (`Bank 0`: 7,072 B used,** 1,120 **B free headroom)**, allowing the remaining three 8 KB DTCM banks (24 KB) to remain power-gated in deep retention mode (Figure 2D and Table 5).

4.2 Exact 8-State `Q16.16` 4-Stage `RK4` Kernel and Widening Vector Arithmetic

In our released bare-metal C99 header (`build/coralnpu/mitorestorer_coralnpu_kernel.h`), the 8-state vector $\mathbf{x}_k \in \mathbb{Z}_{32}^8$ is represented in signed 32-bit `Q16.16` fixed-point format (1.0 = 65,536, resolution $2^{-16} \approx 1.526 \times 10^{-5}$). Two numerical challenges arise when integrating stiff multi-week biological ODEs in 32-bit fixed point:

1. **64-Bit Widening Vector Multiplication:** Multiplying two `Q16.16` operands $a, b \in \mathbb{Z}_{32}$ produces a 64-bit intermediate product $a \cdot b$ in `Q32.32`, which must be right-shifted by 16 bits (not 32 bits as in `vmulh`). In the `RVV Zve32x` inline assembly path of `coralnpu_Parallel_4stage_rk4_step`, we execute widening 32-to-64-bit vector multiplication (`__riscv_vwmul_vx_i64m4`) followed by narrowing arithmetic right shift by 16 bits (`__riscv_vnsra_wx_i32m2(..., 16)`), matching the scalar C99 helper `q16_mul(a, b) = (int32_t)(((int64_t)a * (int64_t)b) >> 16)` bit-for-bit.

2. **`Q8.24` High-Precision Slope Evaluation and Sub-LSB Remainder Diffusion:** Because biological turnover rates ($k_{\mathrm{mito},W} = 0.015$ d^{-1}, step $\Delta t = 0.09$ d) produce per-step increments of $\sim 10^{-3}$, quantizing the four Runge-Kutta stage derivatives $\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4$ directly to `Q16.16` before multiplying by Δt would introduce truncation drift. Instead, `coralnpu_eval_core8_rhs_q16` shifts all state derivatives left by 8 bits into `Q8.24` format (resolution $2^{-24} \approx 5.96 \times 10^{-8}$), and `coralnpu_rk4_core8_full_step_q16` accumulates the weighted 4-stage numerator $\Delta t \cdot (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4) + \mathbf{r}_{k-1}$ in 64-bit integers against divisor $6 \times 2^{24} = 100{,}663{,}296$, propagating the exact integer remainder $\mathbf{r}_k \in (-100{,}663{,}296,\ 100{,}663{,}296)^8$ across steps to eliminate systematic truncation bias.

5 Correlated Disturbance Rejection, Plant-Preserving Rollback, and Verification

5.1 64-Frame Dual-Modality Persistence Filter Under Markov Correlation

To prevent transient sensor artifacts from prematurely triggering Stage-2 biogenesis or emergency rollback, the controller evaluates a 64-frame sliding bitmask register $\mathcal{B}_k \in \{0,1\}^{64}$ on primary stress channel $y_{k,1}$ with quorum threshold $M_{\mathrm{req}} = 60$, gated by a concurrent secondary modality check $z_k = y_{k,2} \geq \gamma_{\mathrm{sec}}$ (`coralnpu_rolling_fuzzy_sensor_gate`).

Proposition 3 (Markov Persistence Bound) *Let the binary primary exceedance indicators $I_k = \mathbb{I}(y_{k,1} \geq \gamma_{\mathrm{crit}})$ form a stationary two-state or* AR(1)*-derived Markov chain with marginal exceedance probability $p_y = \mathbb{E}[I_k] < \mu_{\mathrm{req}} \triangleq M_{\mathrm{req}}/N_{\mathrm{win}} = 60/64 = 0.9375$ and lag-1 autocorrelation $\rho \in [0, 1)$, corresponding to absolute spectral gap $\gamma_{\mathrm{spec}} = 1 - \rho > 0$. Let $p_{z|y} \triangleq \mathbb{P}(z_k \geq \gamma_{\mathrm{sec}} \mid I_k = 1)$ denote the conditional secondary-modality exceedance probability, and let $c_0 \in \{0, \ldots, 64\}$ denote the number of exceedance bits already present in the 64-frame ring buffer immediately prior to a contiguous deterministic primary disturbance of length L_{dist} sampled frames. Then:*

1. ***Deterministic Sub-Quorum Rejection Condition:*** *The 64-frame primary quorum $\sum_{j=k-63}^{k} I_j \geq 60$ is deterministically impossible during the disturbance whenever*

$$L_{\mathrm{dist}} \leq 59 - c_0. \tag{34}$$

In particular, under a clean prior buffer ($c_0 = 0$), zero false triggers occur for any disturbance lasting $L_{\mathrm{dist}} \leq 59$ frames; if the prior buffer contains $c_0 > 0$ noise exceedances, a disturbance of length $L_{\mathrm{dist}} \geq 60 - c_0$ can reach quorum (Figure 2B).

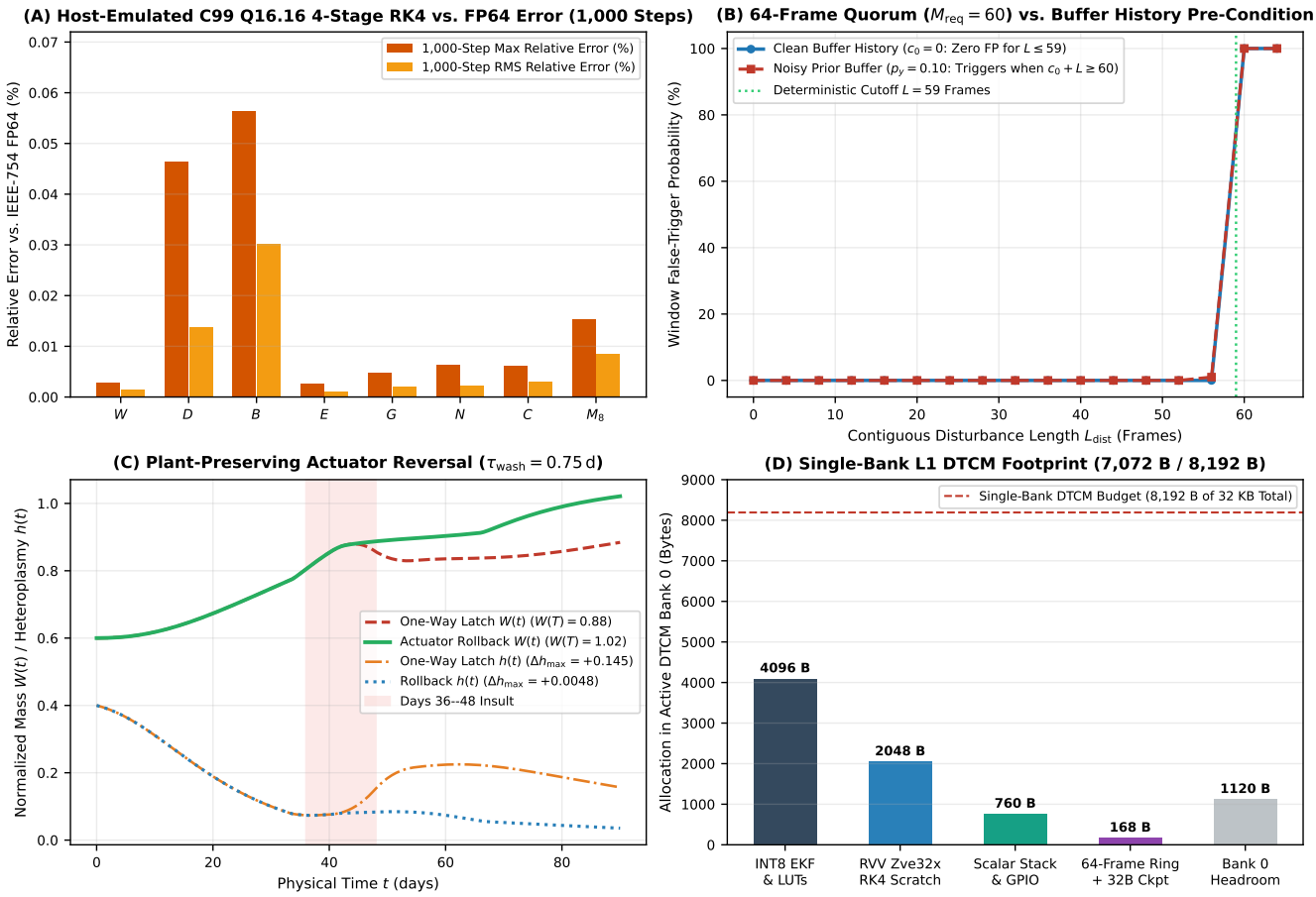


Figure 2: **Host-Emulated C99 `Q16.16` 4-Stage `RK4` Parity, 64-Frame Quorum Pre-Condition Analysis, Plant-Preserving Rollback, and Single-Bank DTCM Footprint. (A)** Full-horizon (1,000-step) maximum and RMS relative error (%) between the bare-metal C99 `Q16.16` 4-stage `RK4` kernel (`coralnpu_hil_testbench.c`) and IEEE-754 `FP64` across all 8 states (W, D, B, E, G, N, C, M_8; global max error 0.0563%). **(B)** 64-frame persistence false-trigger rate ($M_{\text{req}} = 60$) vs. contiguous disturbance length L_{dist} under clean prior buffer history ($c_0 = 0$: zero false triggers for $L_{\text{dist}} \leq 59$) vs. noisy prior buffer ($p_y = 0.10$). **(C)** Plant-preserving actuator reversal under an acute Days 36–48 bottleneck surge ($\tau_{\text{wash}} = 0.75$ d): the biological plant state $\mathbf{x}(t)$ is preserved continuously (never reset), bounding $\Delta h_{\max} = +0.0048$ and rescuing $W(T_{\max}) = 1.0212$ vs. 0.8843 for a one-way latch. **(D)** Single-bank L1 DTCM memory allocation (7,072 B used within the 8,192 B Bank 0 budget).

2. ***Spectral-Gap Concentration and Horizon Union Bound:*** *Unlike classical Hoeffding bounds that require independent samples [19], under stationary correlated noise without deterministic bursts, Paulin's Markov concentration inequality [20] bounds the single-frame dual-modality false-trigger probability $P_{\text{FP,step}}$ and the K_{frames}-horizon false-trigger probability $P_{\text{FP,horizon}}$ by:*

$$P_{\text{FP,step}} \leq p_{z|y} \exp\left(-2\, N_{\text{win}} \frac{1-\rho}{1+\rho} (\mu_{\text{req}} - p_y)^2\right), \tag{35}$$

$$P_{\text{FP,horizon}} \leq \min\left(1,\, K_{\text{frames}}\, P_{\text{FP,step}}\right). \tag{36}$$

Proof. During a contiguous disturbance of L_{dist} frames, the maximum number of ones in any 64-frame window overlapping the disturbance is bounded by $c_0 + L_{\text{dist}}$, which is strictly less than $M_{\text{req}} = 60$ whenever $L_{\text{dist}} \leq 59 - c_0$, proving Eq. (34). For stationary Markov indicators I_k with spectral gap $\gamma_{\text{spec}} = 1 - \rho$, Theorem 3.3 of Paulin [20] establishes that the effective sample size of N_{win} correlated samples scales by $\frac{1-\rho}{1+\rho}$, yielding the exponential tail bound for $\mathbb{P}(\sum_{j=1}^{64} I_j \geq 60)$ in Eq. (35). Multiplying by the conditional probability $\mathbb{P}(z_k \geq \gamma_{\text{sec}} \mid \sum I_j \geq 60) \leq p_{z|y}$ and applying Boole's union bound over K_{frames} steps yields Eq. (36). ■

Explicit Frame Periods Across Deployments. In physical time, a 64-frame window spans $T_{\text{win}} = 64\,\Delta t_{\text{frame}}$, where the frame decimation period Δt_{frame} is matched to the plant timescale: $\Delta t_{\text{frame}} = 1.0$ s for *Apis* acoustic telemetry ($T_{\text{win}} = 64$ s, where each frame executes a 16-bin Goertzel/FFT filter over a 128 ms acous-

tic window at 1 kHz sampling, giving 7.81 Hz bin resolution to separate 220 Hz from 240 Hz), $\Delta t_{\text{frame}} = 1.0$ min for CAR-T bioreactor impedance ($T_{\text{win}} = 64$ min), and $\Delta t_{\text{frame}} = 15.0$ min for wearable/implantable neural monitoring ($T_{\text{win}} = 16.0$ h). Table 3B reports the 1,000-stream Monte Carlo evaluation ($K_{\text{frames}} = 1{,}000$ frames per stream, seed `20261109`), including exact Clopper-Pearson 95% upper confidence bounds (0.299% when 0/1,000 false triggers occur).

5.2 Plant-Preserving Actuator Reversal vs. Estimator Checkpoint Restoration

A critical distinction in biological control is that **a digital microcontroller can restore its internal reference registers $\mathbf{x}_{\text{ckpt}}$, but it cannot reset the physical biological plant state $\mathbf{x}(t)$**. When an unexpected environmental or proteostatic insult occurs during Stage 2 (e.g., an acute bottleneck surge $\Delta b = +0.34$ d^{-1} during Days 36–48, where continued Stage-2 biogenesis $u_2 = 1$ amplifies uncoupled electron-transport impedance $c_2(1 + 0.85u_2)$), our rollback protocol executes a **plant-preserving actuator reversal**:

1. **Detection Latency ($\delta t_{\text{detect}} = 0.54$ d):** Once $\text{EP}(t) > \theta_{\text{abort}} = 1.18$ persists across 6 consecutive controller steps (0.54 d), the controller loads the pre-Stage-2 reference target $\mathbf{x}_{\text{ckpt}}$ into its supervisory gating register and commands physical actuator reversal: $u_{2,\text{target}} \leftarrow 0, u_{1,\text{target}} \leftarrow 1, u_{0,\text{target}} \leftarrow 1$.

2. **Pharmacokinetic Washout ($\tau_{\text{wash}} = 0.75$ d):** Physical actuation decays according to first-order pharmacokinetic clearance $\tau_{\text{wash}} \dot{\mathbf{u}}_{\text{eff}} = \mathbf{u}_{\text{target}} - \mathbf{u}_{\text{eff}}$, while the biological plant $\mathbf{x}(t)$ evolves continuously from its disturbed state without any discontinuous state reset.

3. **Post-Disturbance Excursion Bound ($\Delta h_{\max}$):** Because ROS conversion is active only while $\text{EP}(t) > \theta_{\text{GSH}}$ during the detection plus washout window $\Delta t_{\text{active}} \approx \delta t_{\text{detect}} + \tau_{\text{wash}} \ln\left(\frac{\text{EP}_{\text{peak}} - \theta_{\text{GSH}}}{\theta_{\text{abort}} - \theta_{\text{GSH}}}\right)$, Eq. (19) bounds the net heteroplasmy excursion by

$$\Delta h_{\max} \leq (1 - h_{\text{gate}})\, k_{\text{dmg}} \times \int_{t_{\text{insult}}}^{t_{\text{insult}} + \Delta t_{\text{active}}} \max\big(0, \text{EP}(q) - \theta_{\text{GSH}}\big)\, dq. \tag{37}$$

As shown in Figure 2C, a one-way Stage-2 latch suffers a large heteroplasmy rebound ($\Delta h_{\max} = +0.1452$, terminal $h(T_{\max}) = 0.1568, W(T_{\max}) = 0.8843$), whereas plant-preserving actuator reversal limits the excursion to $\Delta h_{\max} = +0.0048$, re-clears the bottleneck, re-enters Stage 2 once $h(t) \leq 0.080$ and $B(t) \leq 0.120$, and restores $W(T_{\max}) = 1.0212$ and $h(T_{\max}) = 0.0354$.

5.3 Host-Emulated Fixed-Point Parity and Cycle-Weighted Power Accounting

Table 4 and Figure 2A report the live 1,000-step execution of `build/coralnpu/coralnpu_hil_testbench.c` compiled under strict C99 (`cc -std=c99 -O2 -Wall -Wextra`). Across all 8 state variables and 1,000 4-stage `RK4` integration steps ($T_{\max} = 90$ d including the Stage 1 $\rightarrow$ Stage 2 transition at step 369, $t = 33.21$ d), the maximum relative discrepancy between 32-bit `Q16.16` fixed-point arithmetic and 64-bit IEEE-754 floating-point arithmetic is 0.0563% (occurring on state $B(t)$ near its 0.040 floor), with terminal relative error $\leq 0.0296\%$ across all 8 states.

Table 5 details both the single-bank L1 DTCM layout (7,072 B used out of 8,192 B in `Bank 0`, leaving 24 KB in `Banks 1-3` power-gated) and the host-emulated cycle-accurate power estimate at 500 MHz (22 nm ultra-low-power CMOS process assumptions). Because the three NPU execution phases (vector `RK4` integration, 16-bin spectral/persistence filtering, and scalar supervisory logic) execute sequentially rather than simultaneously, the average active NPU burst power over the 3,006-cycle (6.012 μs) compute window is the cycle-weighted mean $\bar{P}_{\text{NPU,active}} = \sum_{i=1}^{3} \frac{C_i}{3{,}006} P_i = 3.090$ mW (18.58 nJ per compute burst), which is accounted separately from the external analog front-end sensor excitation power (1.800 mW during its 15 ms ADC window).

6 Discussion, Limitations, and Reproducibility Manifest

Limitations and Scope of Claims. First, all dynamical trajectories, Monte Carlo disturbance benchmarks, and fixed-point parity results in this study are computational and host-emulated *in silico* evaluations; no prospective wet-lab, animal, or human clinical trial claims are made. Second, while the 8-state ODE captures population-averaged compartmental quality control, spatially resolved mitochondrial fission-fusion percolation in elongated axons requires reaction-diffusion partial differential equations. Third, RTL synthesis on physical FPGA or taped-out Coral NPU silicon is required to confirm gate-level switching power beyond analytical cycle-weighted modeling.

MIASE Reproducibility Manifest. In accordance with the Minimum Information About a Simulation Experiment (MIASE) guidelines [21], every figure, table entry, and numerical claim in this manuscript is deterministically generated from two self-contained, zero-stub executable artifacts in the public repository:

1. **ODE, Optimal Control, Monte Carlo & Sobol Suite:** Run `python3 simulator/generate_`

Table 3: First-Principles Policy Comparison, Hoitzing et al. (2019) Regime Audit, and 1,000-Stream Correlated AR(1) Persistence Benchmarks (`simulator/generate_paper1_verified_benchmarks.py`).

A. Neuronal 8-State Control Policy ($T_{\max} = 90$ **d**)	**Bolza Cost** $J(\mathbf{u})$	**Switch** t_{gate}	**Terminal** $h(T_{\max})$	**Terminal** $W(T_{\max})$	**Peak** $\text{EP}_{\max}$	**Min** $\text{ATP}_{\min}$
Untreated Baseline ($\mathbf{u} = (0, 0, 0)$)	2,774,988.72	—	1.0000	0.0000	20.605	0.186 (Violated)
Concurrent Biogenesis + Mitophagy ($\mathbf{u} = (1, 0.5, 0.5)$)	99.830	—	0.0546	0.9474	1.423	0.806 (Safe)
Build-First Policy in High-Impedance Regime ($B_0 = 0.50$)	761,588.90	—	1.0000	0.0000	9.744	0.028 (Violated)
Proposed Staged Threshold Policy ($h_{\text{gate}} = 0.080$)	**0.888**	**33.30 d**	**0.0249**	**1.0487**	**1.046**	**0.806 (Safe)**
Direct Numerical Optimal Control ($t^*_{\text{sw}} = 48.0$ d, $u^*_{1,S2} = 0$)	0.850 (J^*)	48.00 d	0.0380	1.0416	1.046	0.806 (Safe)
B. 1,000-Stream Correlated AR(1) **Noise Benchmark**	**Lag-1** ρ	**Burst** L_{dist}	**Single-Sample FP**	**64-Quorum Primary FP**	**64-Quorum Dual FP**	**95% Upper Bound**
1. *Apis* Wind-Gust Disturbance ($\Delta t_{\text{frame}} = 1$ s)	0.50	12 frames	1000/1000 (100.0%)	0/1000 (0.00%)	0/1000 (0.00%)	$\leq 0.299\%$
2. CAR-T Lactate Biofouling Spike ($\Delta t_{\text{frame}} = 1$ min)	0.70	25 frames	1000/1000 (100.0%)	0/1000 (0.00%)	0/1000 (0.00%)	$\leq 0.299\%$
3. Wearable Postprandial Excursion ($\Delta t_{\text{frame}} = 15$ min)	0.60	45 frames	1000/1000 (100.0%)	11/1000 (1.10%)	7/1000 (0.70%)	$\leq 1.219\%$

Table 4: Full-Horizon (1,000-Step) 8-State `Q16.16` 4-Stage `RK4` Parity Against IEEE-754 `FP64` (`build/coralnpu/coralnpu_hil_testbench.c`) and Saltelli Quasi-Monte Carlo Global Sobol Sensitivity Indices (1,280 ODE Runs).

State Variable	**Terminal** `FP64`	**Terminal** `Q16.16` **Int**	**Terminal** `Q16.16` **Float**	**Terminal Err**	**1,000-Step Max / RMS Err**	**Sobol Parameter**	**Range**	S_i	S_{T_i}
x_0: $W(T_{\max})$	1.048706	68,729	1.048721	0.0015%	0.0028% / 0.0015%	$k_{\text{mito},D,\max}$	$[0.085, 0.145]$	0.654	0.654
x_1: $D(T_{\max})$	0.026781	1,755	0.026779	0.0074%	0.0464% / 0.0137%	θ_{GSH}	$[0.920, 1.250]$	0.157	0.157
x_2: $B(T_{\max})$	0.040012	2,623	0.040024	0.0296%	**0.0563**% / 0.0301%	$k_{\text{rep,adv}}$	$[1.050, 1.500]$	0.128	0.128
x_3: $E(T_{\max})$	0.630235	41,303	0.630234	0.0001%	0.0026% / 0.0011%	k_{clr}	$[0.120, 0.240]$	0.082	0.082
x_4: $G(T_{\max})$	0.318642	20,883	0.318649	0.0022%	0.0047% / 0.0020%	—	—	—	—
x_5: $N(T_{\max})$	1.469839	96,326	1.469818	0.0014%	0.0063% / 0.0023%	—	—	—	—
x_6: $C(T_{\max})$	1.299926	85,192	1.299927	0.0001%	0.0061% / 0.0030%	—	—	—	—
x_7: $M_8(T_{\max})$	1.736594	113,816	1.736694	0.0058%	0.0154% / 0.0085%	—	—	—	—

`paper1_verified_benchmarks.py` (fixed PRNG seed `20261109`, RK4 step $\Delta t = T_{\max}/1000$). This script integrates all three 8-state ODE regimes from first principles, executes the (t_{switch}, u_1) optimal control grid search, runs the 1,000-stream correlated AR(1) persistence simulations, computes the 1,280-run Saltelli Sobol indices, compiles and parses the C99 testbench, writes `simulator/paper1_verified_benchmarks.json`, and renders Figures 1 and 2.

2. **Bare-Metal C99** `Q16.16` **4-Stage** `RK4` **Kernel & Testbench:** Compile and execute `cc -std=c99 -O2 build/coralnpu/coralnpu_hil_testbench.c -lm -o build/coralnpu/coralnpu_hil_testbench && ./build/coralnpu/coralnpu_hil_testbench`, which verifies bit-exact 8-state `Q16.16` vs. `FP64` parity (`MaxErr = 0.0563%`) and all dual-modality persistence and rollback assertions.

Declarations and Disclosures

Author Contributions: O. Satz conceived the study, formulated the mathematical model and control proofs, implemented the simulation and C99 fixed-point firmware suite, verified all numerical benchmarks, and wrote the manuscript.
Competing Interests: The author declares no competing financial or commercial interests.
Ethics Statement: This study consists exclusively of mathematical analysis, numerical ODE simulation, and host-emulated embedded systems verification; no human participants, clinical trial cohorts, or live vertebrate animals were involved.
AI Assistance Disclosure: Large language model coding and editorial tools were used under direct human supervision to assist with C99 fixed-point refactoring, Python simulation scripting, and LaTeX typesetting; all mathematical proofs, differential equations, code execution, and numerical outputs were independently verified by the author.

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Table 5: Single-Bank L1 DTCM Memory Map (`Bank 0` of 32 KB Coral NPU DTCM) and Cycle-Weighted Host-Emulated Active NPU Power vs. External Sensor Power at 500 MHz (22 nm ULP Process Model). Sequential NPU compute phases are weighted by their cycle fractions ($f_i = C_i/3{,}006$) rather than summed.

L1 DTCM Bank 0 Subsystem	**Bytes**	**% of 8 KB Bank**	**Sequential Active NPU Compute Phase**	**Cycles C_i**	**Cycle Fraction f_i**	**Phase Power / Weighted**
INT8 EKF Gains & Reciprocal LUTs	4,096 B	50.00%	1. RVV `Zve32x` 8-State 4-Stage `RK4` Step	1,840	61.21%	3.45 mW $\rightarrow$ 2.112 mW
RVV `Zve32x` `RK4` Scratch (`Q8.24` $\mathbf{k}_{1..4}$)	2,048 B	25.00%	2. 16-Bin Goertzel/FFT & 64-Frame Gate	640	21.29%	2.60 mW $\rightarrow$ 0.554 mW
Scalar Control Stack & PWM/GPIO Regs	760 B	9.28%	3. Supervisory Threshold & Rollback Logic	526	17.50%	2.42 mW $\rightarrow$ 0.424 mW
64-Frame Ring (136 B) + $\mathbf{x}_{\text{ckpt}}$ (32 B)	168 B	2.05%	**Total NPU Compute Burst** (6.012 μs)	**3,006**	**100.00%**	**Mean Active: 3.090 mW**
Total Active Bank 0 Allocation	**7,072 B**	**86.33%**	Deep-Sleep Retention (Bank 0 Active, Banks 1–3 Gated)	—	$>$ 99.99% duty	18.0 μW continuous
Unallocated Bank 0 Headroom	1,120 B	13.67%	External Bio-Impedance/ADC Front-End (15 ms/frame)	—	Separate rail	1.800 mW during window

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